

assignment-02-code

August 28, 2026

1 Assignment II

1.1 Import External Libraries

```
[1]: import numpy as np

from sklearn.preprocessing import normalize
```

1.2 2. Normalization Using Python

```
[2]: v1 = np.array([[2], [1]])
print(f"Original vector:\n{v1}")

print()

v2 = np.round(v1 / np.linalg.norm(v1), 4)
print(f"Normalized vector:\n{v2}")
```

Original vector:

```
[[2]
 [1]]
```

Normalized vector:

```
[[0.8944]
 [0.4472]]
```

1.3 4. Normalization Using Python with Complex Coefficient

```
[3]: v1 = np.array([[2], [1+1j]])
v2 = np.round(v1 / np.linalg.norm(v1), 4)
print(f"Normalized vector:\n{v2}")
```

Normalized vector:
[[0.8165+0.j]
 [0.4082+0.4082j]]

1.4 7. Vector Normalization by Hand and Using Python

```
[4]: # v1 = (1, 2) in the |0>/|1> basis
v1_01 = np.array([[1], [2]])

# basis change matrix from |0>/|1> to |+>/|-> coefficients
M = (1 / np.sqrt(2)) * np.array([[1, 1], [1, -1]])

v1_pm = np.round(M @ v1_01, 4)
print("v1 in +/- basis (unnormalized):", v1_pm.ravel())

v1_pm_normalized = np.round(v1_pm / np.linalg.norm(v1_pm), 4)
print("v1 in +/- basis (normalized):", v1_pm_normalized.ravel())
```

v1 in +/- basis (unnormalized): [2.1213 -0.7071]
v1 in +/- basis (normalized): [0.9487 -0.3162]

1.5 9. Vector Normalization by Hand and Using Python 2

```
[5]: v1 = np.array([[1], [2]])
v1_normalized = v1 / np.linalg.norm(v1)
print("v1 normalized in 0/1 basis:", v1_normalized.ravel())

prob_0 = np.abs(v1_normalized[0, 0]) ** 2
print(f"P(|0>): {prob_0:.4f}")
```

v1 normalized in 0/1 basis: [0.4472136 0.89442719]
P(|0>): 0.2000
